

EXAM 1 Study Guide

Format

57 questions + Extra Credit question (short-answer)

Multiple Choice – 28 Qs

Matching – 4 Qs

True/False – 12Qs

Fill in the blank – 10 Qs

Multiple answers – 3

Lectures 1-2: Neural Cells of the Nervous System

- Know and be able to identify the distinct components of a neuron, for example:
 - Soma
 - Neurites
 - Dendrites
 - Axons
 - Membrane
 - Endoplasmic reticulum,
 - Nucleus
- Know the different neuron types and their properties
 - Structural: unipolar, multipolar, bipolar, pseudo-unipolar
 - Functional: interneurons, motor neurons, sensory neurons
- Know if an neuron/axon is an afferent or efferent
 - Efferent – information exits area
 - Afferent – information arrives in area
- Know the different types of glia cells & their major properties/functions

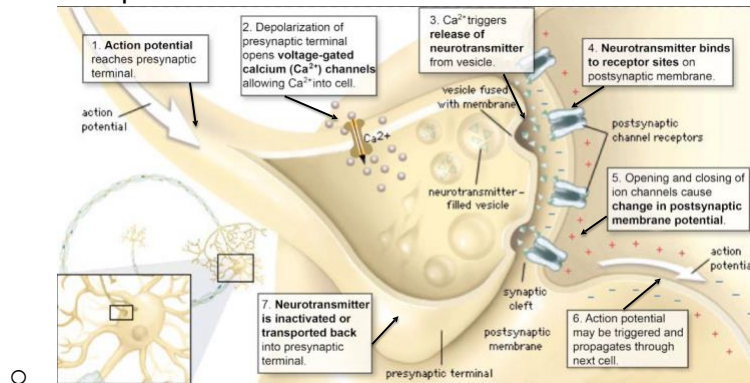
Lecture 3 AP

- Know all the required information for the paper
 - General properties of AP
 - Stimulus
 - Threshold
 - Axon hillock
 - All-or-none
 - Ion channels (gated & non-gated)
 - RMP properties, for example:
 - Allows neuron to respond rapidly to a stimulus
 - Determined by ionic concentrations
 - Na/K⁺ pump helps maintain it
 - How myelin affects AP propagation
 - Direction of ion movement during AP
 - Active versus passive forces involved in flow of ions

- Phases of AP and refractory periods

Lecture 4 Synapses

- Different types and their properties
 - Chemical & electrical
 - Local graded potentials
 - Differences with AP
 - Ions involved in EPSPs & IPSPs
 - Temporal & Spatial Summation
- General steps that lead from AP to neurotransmitter release



Lecture 5 Neurotransmitters

- Know how they exert their effects at different receptors
 - 2 types of postsynaptic receptors
 - Ligand-gated ion channels
 - GPCRs
- Know the sources of their synthesis
- General properties of exogenous drugs
 - Stimulants e.g. cocaine
 - Depressants e.g. opioids
- PHARMACODYNAMICS
 - Agonist
 - Antagonist
 - Direct or indirect; e.g. Cocaine is an indirect agonist
 - Affinity & efficacy
- Ways to clear neurotransmitters

Lecture 6 Nervous System

- Properties of major subdivisions; eg
 - PNS
 - Somatic
 - Controls movement via skeletal muscles
 - Autonomic
 - Controls involuntary movements

- Sympathetic – flight or fight; affects HR, etc
- Parasympathetic – rest & digest
- Spinal nerves
 - 31 pairs
 - **Dorsal – Afferent**
 - **Ventral – Efferent**
 - = **DAVE's** spine
- CNS
 - Be able to identify brain in Directional terms
 - Coronal, Horizontal, Sagittal
 - Medial – Lateral, Anterior-Posterior, Dorsal- Ventral, Proximal – Distal, Ipsilateral-Contralateral
 - Know major brain divisions
 - Hindbrain: medulla, pons, cerebellum
 - Be able to identify 4 brain lobes (forebrain)
 - Know function of forebrain subcortical regions

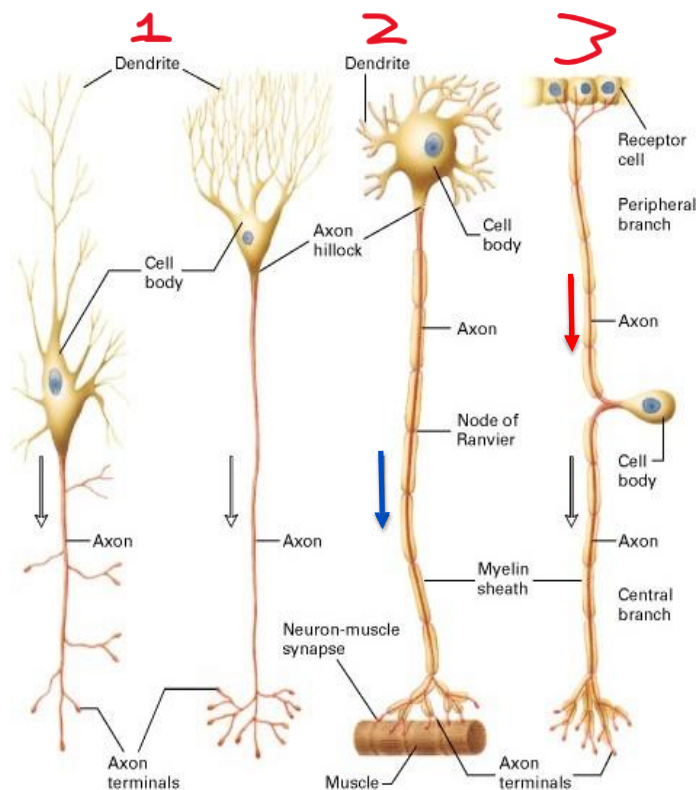
Example questions:

Multiple choice: Which glial cell(s) connects to capillaries?

- A. Astrocytes
 B. Microglia
 C. Oligodendrocytes
 D. A & C
 E. All of the above

Matching: Match the neurons in the image below with their correct type:

- A – interneuron: 1
 B – motor neuron: 2
 C – sensory neuron: 3



Fill in the blank: During an action potential, K^+ flows ____ and Na^+ flows ____ of the axon at the ____.

out, in, nodes of Ranvier

True or False: MAO inhibitors are indirect agonists. True/False